

Chapter 5: Probability

A Level Statistics Notes · Rexponential

Why this chapter matters

Probability is the mathematical backbone the rest of the course builds on — the binomial and Normal distributions in Chapter 6, and every hypothesis test in Chapter 7, are all built from the ideas in this chapter. Venn diagrams, tree diagrams, and the rules for combining probabilities are worth mastering thoroughly, since a shaky grasp here makes every later chapter harder than it needs to be.

5.1 Calculating probabilities and Venn diagrams

Concept

Probability measures how likely an event is, on a scale from 0 (impossible) to 1 (certain).

A **Venn diagram** represents events as overlapping regions inside a rectangle standing for the whole **sample space**. It makes it easy to visualise and calculate probabilities involving unions, intersections and complements (see the vocabulary table below).

When two events can happen at the same time, simply adding their individual probabilities counts the overlap twice. The addition rule below corrects for this by subtracting that overlap once.

Key vocabulary

Term	Meaning
Sample space	The set of all possible outcomes
Event	A subset of the sample space
Union, $A \cup B$	A or B (or both) occur
Intersection, $A \cap B$	Both A and B occur
Complement, A'	A does not occur

Key formula

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Procedure

1. Draw a rectangle for the sample space and a circle for each event.
2. Fill in the intersection region(s) first, then work outwards to the regions belonging to only one event, then the region outside all circles.
3. Use $P(A) + P(B) - P(A \cap B)$ whenever combining probabilities with "or", to avoid double-counting the overlap.
4. Read required probabilities directly off the completed diagram, or calculate them algebraically using the addition rule.

Worked example

EXAMPLE 5.1

In a class of 30 students, 18 study French, 15 study Spanish, and 8 study both. (a) Draw a Venn diagram to represent this information. (b) Find the probability that a randomly chosen student studies French or Spanish (or both). (c) Find the probability that a student studies neither language.

Step 1 (part a). French only (study French but not Spanish): $18 - 8 = 10$. Spanish only (study Spanish but not French): $15 - 8 = 7$. Both: 8 (given). Neither (study neither language): $30 - (10 + 7 + 8) = 5$. (M1 A1)

Step 2 (part b). Using the addition rule (this should agree with summing the Venn diagram regions from Step 1: $10 + 7 + 8 = 25$ out of 30 students): (M1)

$$P(F \cup S) = \frac{18}{30} + \frac{15}{30} - \frac{8}{30} = \frac{25}{30} = \frac{5}{6}$$

(A1)

Step 3 (part c). $P(\text{neither}) = 1 - P(F \cup S) = 1 - 5/6 = 1/6$, matching the 5 students found in Step 1 ($5/30 = 1/6$). (A1)

Answer (b) $5/6$. (c) $1/6$.

EXAM TIP

Always fill a Venn diagram from the *inside out*, starting with the intersection — trying to place the "French only" and "Spanish only" regions first is the most common source of arithmetic slips, since those numbers depend on the overlap already being known.

5.2 Mutually exclusive and independent events

Concept

Two events are **mutually exclusive** if they cannot both happen at the same time (their intersection is empty, so $P(A \cap B) = 0$), which simplifies the addition rule. Two events are **independent** if one occurring has no effect on the probability of the other — this is a different idea from being mutually exclusive, and it is easy to mix the two up.

In fact, two events with non-zero probabilities cannot be both mutually exclusive and independent at the same time. This is because mutually exclusive events give $P(A \cap B) = 0$, whereas independent events give $P(A \cap B) = P(A) \times P(B)$, which is greater than zero whenever both $P(A)$ and $P(B)$ are greater than zero — the two conditions contradict each other unless one event is impossible.

Key vocabulary

Term	Meaning
Mutually exclusive	Events that cannot occur together: $P(A \cap B) = 0$
Independent	Events where one occurring does not affect the probability of the other

Key formulas

For mutually exclusive events:

$$P(A \cup B) = P(A) + P(B)$$

For independent events:

$$P(A \cap B) = P(A) \times P(B)$$

Procedure

1. To test whether two events are mutually exclusive, check whether $P(A \cap B) = 0$ (can they physically happen together?).
2. To test whether two events are independent, check whether $P(A \cap B) = P(A) \times P(B)$ using the given or calculated probabilities — if the two sides do not match, the events are not independent.
3. Use the simplified addition rule only once mutual exclusivity is confirmed; use the multiplication rule only once independence is confirmed (or assumed, if the question states it).

Worked example

EXAMPLE 5.2

For two events A and B, $P(A) = 0.4$, $P(B) = 0.5$, and $P(A \cap B) = 0.2$. Determine whether A and B are (a) mutually exclusive, (b) independent.

Step 1 (part a). Since $P(A \cap B) = 0.2 \neq 0$, A and B are **not** mutually exclusive — they can occur together. (B1)

Step 2 (part b). Check $P(A) \times P(B)$ against $P(A \cap B)$. (M1)

$$P(A) \times P(B) = 0.4 \times 0.5 = 0.2$$

Since this equals the given $P(A \cap B) = 0.2$, A and B **are** independent. (A1)

Answer (a) Not mutually exclusive. (b) Independent.

EXAM TIP

Students very often assume "independent" and "mutually exclusive" are opposites or interchangeable. A concrete counter-example: rolling a die and getting an even number vs. getting a number greater than 4 are neither mutually exclusive (6 satisfies both) nor independent — use numeric checks like the one above rather than intuition alone.

5.3 Tree diagrams

Concept

A **tree diagram** shows a sequence of events as branches, with the probability of each branch written on it. Along any single path, probabilities are **multiplied**; when there is more than one path leading to the same outcome, those path probabilities are **added**.

Tree diagrams are especially useful when events are not independent, since later branch probabilities can change depending on what happened earlier (sampling without replacement).

Key vocabulary

Term	Meaning
Tree diagram	A diagram showing possible sequences of events as branches with probabilities
With replacement	An item is returned before the next selection, so probabilities stay the same
Without replacement	An item is not returned, so later probabilities change

Key formula

$$P(A \text{ then } B) = P(A) \times P(B \mid A)$$

Here, $P(B \mid A)$ is the probability of the second branch's outcome given that A (the first branch's outcome) has already happened — this is a conditional probability, which Section 5.4 explains fully.

Procedure

1. Draw each stage as a set of branches, labelling every branch with its probability — branches from the same point must sum to 1.
2. For "without replacement" problems, adjust the probabilities on later branches to reflect what was removed at the earlier stage.
3. Multiply along a path to find the probability of that specific sequence of outcomes.
4. If more than one path gives the required outcome, add the probabilities of those paths together.

Worked example

EXAMPLE 5.3

A bag contains 5 red and 3 blue counters. Two counters are drawn at random, without replacement. Find the probability that both counters are the same colour.

Step 1. Draw the tree: first draw $P(\text{Red}) = 5/8$, $P(\text{Blue}) = 3/8$. After removing one counter, second-draw probabilities change. (M1)

Step 2. After one red counter is removed, 4 red and 3 blue counters remain (7 in total). Both red: (M1)

$$P(RR) = \frac{5}{8} \times \frac{4}{7} = \frac{20}{56}$$

After one blue counter is removed, 5 red and 2 blue counters remain (7 in total). Both blue:

$$P(BB) = \frac{3}{8} \times \frac{2}{7} = \frac{6}{56}$$

(A1 A1)

Step 3. Add the two paths, since either gives "same colour". (M1)

$$P(\text{same colour}) = \frac{20}{56} + \frac{6}{56} = \frac{26}{56} = \frac{13}{28}$$

(A1)

Answer $P(\text{same colour}) = 13/28$.

EXAM TIP

The most common slip is forgetting to adjust the denominator (and sometimes the numerator) on the second set of branches for "without replacement" questions — state the new total remaining before writing each second-stage probability, rather than copying the first-stage fractions across.

5.4 Conditional probability and set notation

Concept

Conditional probability, written $P(A | B)$, is the probability that A occurs *given that* B has already occurred — it restricts the sample space to only the outcomes where B happened.

This formalises what a tree diagram's later branches already represent, and connects directly back to the independence test in 5.2: if $P(A | B) = P(A)$, then knowing B happened tells you nothing new about A, so A and B are independent.

Key vocabulary

Term	Meaning
$P(A B)$	The probability of A, given that B has occurred
A'	The complement of A (A does not occur)

Key formula

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

Procedure

1. Identify which event is the "given" condition (it goes in the denominator, restricting the sample space).
2. Find $P(A \cap B)$ — from a Venn diagram, a tree diagram, or a two-way table, whichever the question provides.
3. Divide by $P(B)$ (the probability of the condition) to find $P(A | B)$.
4. To test independence using conditional probability, compare $P(A | B)$ to $P(A)$: if they are equal, the events are independent.

Worked example

EXAMPLE 5.4

In a survey, $P(\text{owns a car}) = 0.7$, $P(\text{owns a bike}) = 0.4$, and $P(\text{owns both a car and a bike}) = 0.3$. Find the probability that a randomly chosen person owns a bike, given that they own a car.

Step 1. Identify the condition: "given that they own a car" means car is the condition, so it goes in the denominator. (M1)

Step 2. Substitute the given values into the formula:

$$P(\text{Bike} | \text{Car}) = \frac{P(\text{Bike} \cap \text{Car})}{P(\text{Car})} = \frac{0.3}{0.7} = \frac{3}{7}$$

(A1)

Answer $P(\text{Bike} | \text{Car}) = 3/7 \approx 0.429$.

EXAM TIP

The most common error is putting the wrong probability in the denominator — read the condition carefully: " $P(A | B)$ " is always "probability of A, given B", so B's probability is the denominator, not A's. Get into the habit of writing the formula with the letters filled in (as above) before substituting numbers, rather than guessing which value goes on top.

Chapter summary

Idea	Key point
Addition rule	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Mutually exclusive	$P(A \cap B) = 0$, so $P(A \cup B) = P(A) + P(B)$
Independent	$P(A \cap B) = P(A) \times P(B)$; equivalently $P(A B) = P(A)$
Tree diagrams	Multiply along a path, add across separate paths to the same outcome

Idea	Key point
Conditional probability	$P(A B) = P(A \cap B) / P(B)$

Common exam mistakes

- Filling a Venn diagram from the outside in rather than starting with the intersection.
- Treating "mutually exclusive" and "independent" as the same idea, or as opposites.
- Forgetting to adjust probabilities on later tree branches for "without replacement" scenarios.
- Putting the wrong event's probability in the denominator of a conditional probability calculation.

You should now be able to

- Use Venn diagrams and the addition rule to find probabilities of unions, intersections and complements.
- Test whether two events are mutually exclusive and/or independent, using probability calculations rather than intuition.
- Draw and use tree diagrams, including for sampling without replacement.
- Calculate and interpret conditional probabilities.